Aadi Gupta

B.Tech Student & Developer

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Education

2026–2030 **B.Tech (Class of 2030)**, *Vellore Institute of Technology (VIT)*, Vellore
Focus: Software Engineering, Embedded Systems, and Hardware Prototyping.

2024–2026 **Higher Secondary (Class 12 - CBSE)**, *James Convent School*, Nangloi, Delhi
Developed a strong analytical and problem-solving foundation.

–2024 **Secondary School (Class 10)**, *R.E.D. Senior Secondary School*, Jhajjar, Haryana

Technical Skills

Languages & Web: Python, C/C++, HTML5, CSS3, React.js (Exploring)

Hardware: Arduino, ESP32, Sensor Integration (Ultrasonic, Barometric, Accelerometer), Hardware Prototyping

Tools & Software: Git, GitHub, OrCAD X, SolidWorks, Streamlit

Concepts: IoT Architecture, Edge AI / TinyML, Multi-Agent Systems, B2B SaaS

Projects & Hackathons

2026 **ESP32 Aviation GPWS & Auto-Flap Landing System**, *Sensored BeyondX Hackathon*, VIT Vellore

- **Award:** Secured 2nd Position (as freshers) with Team 'Fantastic Four' in the TEC club hackathon.
- **Hardware Engineering:** Developed a dual-sensor altitude fusion system using BMP180 (barometric) and HC-SR04 (ultrasonic) to prevent CFIT (Controlled Flight Into Terrain) crashes.
- **Embedded Logic:** Programmed an ESP32 in C/C++ to read ADXL345 accelerometer pitch data, actuating servos for automated wing flaps and triggering emergency hazard alarms based on flight altitude and tilt parameters.

2026 **Flowgrid (Healthcare Resource SaaS)**, *ADG Hackgrid*, VIT Vellore

- **Development:** Built a voice-activated web application to help hospital staff quickly coordinate beds, doctors, and equipment during emergencies.
- **AI & Backend:** Implemented the Python and Streamlit backend, integrating the 'faster-whisper' speech-to-text AI model to process clinical requirements.
- **System Architecture:** Engineered a multi-agent backend structure to automate compatibility matching between patient needs and a simulated hospital resource database.

Previous **SAVER Rover Integration**, *Robotics Project*

- **Robotics:** Built an autonomous environmental navigation rover using Arduino C++ for core control logic and decision-making.
- **Sensor Modification:** Engineered a custom directional sonar scanning system using an SR04 sensor attached to a servo motor for obstacle avoidance.
- **Assembly:** Managed end-to-end hardware assembly, wiring, power distribution, and chassis construction.